supervision remain essential to ensure safety, reliability, and ethical compliance. Effective human–drone collaboration depends on intuitive interfaces, real-time communication, and mutual understanding between the operator and the autonomous system.

The goal of HDI design is to create systems where humans and drones can cooperate seamlessly, with drones interpreting operator intent accurately and humans understanding the drone's decisions, limitations, and feedback. However, challenges such as control complexity, interface overload, latency, and operator fatigue continue to limit optimal collaboration. Overcoming these challenges requires the development of user-centered interaction frameworks, adaptive control interfaces, and AI systems that can communicate with humans transparently and contextually.

7.10.1 Developing intuitive human–drone interaction systems

Intuitive human–drone interaction (HDI) systems are key to enabling effective and safe collaboration between operators and drones. These systems must bridge the gap between human cognition and machine autonomy, ensuring that communication is natural, responsive, and error-tolerant. Traditional drone control interfaces rely heavily on manual joysticks or tablet-based applications, which can be cognitively demanding in complex missions.

Recent advancements are introducing multimodal interaction methods, including gesture recognition, voice commands, eye-tracking, and augmented reality (AR) interfaces. These technologies allow operators to control drones intuitively without requiring constant manual input, making operations more efficient and immersive. For instance, AR headsets can display flight data, environmental maps, and drone perspectives in real time, allowing users to make informed decisions quickly.